H.W-3

problem-1

→ Recast the cantelever beam problem as a variational problem. clearly state the test and trial function spaces.

→ **Strong form:**

$$\left.\begin{aligned} u^{(iv)} &= f(x), \quad x \in (0,1) \\ u(0)=0 &, \; u'(0)=0, \text{ clamped end} \\ u''(1)=0 &, \; u'''(1)=0, \text{ (free end)} \end{aligned}\right\} \quad \text{—(1)}$$

step-1 multiply by v and integrate by parts twice.

$$\Rightarrow \int_0^1 v\, u^{(iv)}\, dx = \int_0^1 f\, v\, dx$$

$$\Rightarrow [v u''']_0^1 - \int_0^1 v' u'''\, dx = \int_0^1 f v\, dx$$

$$\Rightarrow -[v' u'']_0^1 + \int_0^1 v'' u''\, dx$$

$$\Rightarrow [u v''']_0^1 - [v' u'']_0^1 + \int_0^1 v'' u''\, dx = \int_0^1 f v\, dx.$$

($u(0)=0 \Rightarrow v(0)=0$; $u''(1)=0 \Rightarrow v''(1)=0$)

$$\therefore \int_0^1 v'' u''\, dx = \int_0^1 f v\, dx \quad \rightarrow \text{weak form} \quad \text{—(2)}$$

↳ it ask for v'', u'' to be finite.

$$\text{So, } V = \left\{ v : \int_0^1 [v^2 + (v')^2 + (v'')^2]\, dx < \alpha, \; v(0)=0, \; v'(0)=0 \right\} \quad \text{—(3)}$$

↑ test space ↑ trial space

↳ which is also test space

They coincide because essential BCs are homogeneous.

Variational form: Find $u \in V$ such that (2)

$$\int_0^1 u''v''\,dx = \int_0^1 fv\,dx \quad \text{for } \forall\ v \in V$$

with V given by 3, trial & test space is V

(2) int interpret the variational problem as an energy minimization problem, clearly stating the energy functional. prob e that variational problem and energy minimization problems are equivalent.

Solu

energy functional for beam version:

$$F(w) = \frac{1}{2}\int_0^1 (w'')^2\,dx - \int_0^1 fw\,dx \qquad \text{——— (4)}$$

Minimization problem: Find $u \in V$ such that $F(u) \leq F(w)$ for all $w \in V$ with V from (3) and F from (4)

vari → min

→ V is (3) is a vector space. so sum and scalar multiple stays inside. $u, w \in V$ then $v := w - u \in V$ and if $u, v \in V$ then $u + \epsilon v \in V$ for some $\epsilon \in \mathbb{R}$

→ $w = u + v$

Now, $F(w) = F(u+v)$

$$= \frac{1}{2}\int_0^1 (u'')^2\,dx + \int_0^1 u''v''\,dx + \frac{1}{2}\int_0^1 (v'')^2\,dx - \int_0^1 fu\,dx - \int_0^1 fv\,dx$$

$$= \underbrace{\frac{1}{2}\int_0^1 (u'')^2\,dx - \int_0^1 fu\,dx}_{= F(u)} + \frac{1}{2}\int_0^1 (v'')^2\,dx + \underbrace{\int_0^1 u''v''\,dx - \int_0^1 fv\,dx}_{= 0 \text{ by } (2)}$$

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So, $F(w) = F(u) + \frac{1}{2}\int_0^1 (v'')^2 dx$ ——— (5)

$\int_0^1 (v'')^2 dx \rightarrow > 0$ since square.

So, $F(w) \geq F(u)$

$\therefore F(u) \leq F(w) \rightarrow$ proved

Now Min. Prob. $\Longrightarrow$ Vari. prob

Let $u \in V$ satisfy $F(u) \leq F(w)$ for all $w \in V$. Fix an arbitrary $v \in V$ and define one variable function:

$$g(\epsilon) = F(u + \epsilon v), \quad \epsilon \in \mathbb{R} \quad \cdots (6)$$

$\rightarrow u + \epsilon v \in V$ for every ϵ, so $g(\epsilon) \geq F(u) = g(0)$

for every ϵ: g has global minimum at $\epsilon = 0$.

Now,

$$g(\epsilon) = \underbrace{\frac{1}{2}\int_0^1 (u'')^2 dx - \int_0^1 fu\,dx}_{\text{constant}} + \epsilon\underbrace{\left[\int_0^1 u''v''dx - \int_0^1 fv\,dx\right]}_{=:c_1} + \frac{\epsilon^2}{2}\underbrace{\int_0^1 (v'')^2 dx}_{=:c_2} \quad \cdots (7)$$

So, g is quadratic polynomial at ϵ, with $c_2 \geq 0$.

Now, $g'(\epsilon) = \underbrace{\int_0^1 u''v''dx - \int_0^1 fv\,dx}_{c_1} + \epsilon\underbrace{\int_0^1 (v'')^2 dx}_{c_2}$

$\Rightarrow g'(\epsilon) = c_1 + \epsilon c_2$

$\rightarrow g$ is differentiable and has an interior global minimum at $\epsilon = 0$, hence $g'(0) = 0$ ⊙

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So, $c_1 = \int_0^1 u'' v'' dx - \int_0^1 fv\,dx = 0$

$\Rightarrow \int_0^1 u'' v'' dx = \int_0^1 fv\,dx$

which is the variational problem.

3

Develop a CG(3) finite element method for this problem.

Sol

→ The method must be conforming $V_h \subset V$. The only smoothness V demand from vari. form is that $\int_0^1 (v'')\,dx < \alpha$.

→ v, v' must be continuous. v'' may jump (kink is allowed)

→ $v \in C^1$

→ Continuity of v and v' is two conditions at each ~~ea~~ ~~end~~ end of an element → so 4 condition per element ↔ a polynomial with 4 free co-efficients

i.e → cubic — Hence CG(3)

→ cubic is the smallest polynomial that works

Mesh and which dof survive:

$h = \frac{1}{M+1}$ ~~M = number of~~ interior nodes.

→ uniform mesh of M elements:

$0 = x_0 < x_1 < x_2 < \cdots < x_{M+1} = 1$ where, $h = x_j - x_{j-1}$

$x_j = jh$.

→ each node carries two dof. → the deflection ξ_j

→ the slope — η_j

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→ clamped end $x_0 = 0$: $v(0) = v'(0) = 0$ → essential.
→ so both dof at node 0 are deleted.

→ Free end $x_{M+1} = 1$: nothing is imposed. both dof at node $M+1$ are kept.

→ Hence

$$V_h = \{ v \in V : v|_{[x_{j-1}, x_j]} = a_j + b_j \cdot x + c_j x^2 + d_j x^3 \}$$

$$= \text{span}\{ \phi_1^1, \dots \phi_{M+1}^1, \phi_1^2 \dots \phi_{M+1}^2 \}$$

$$\dim V_h = 2(M+1)$$

The basis

→ two shapes per node, fixed by nodal table

basis or trial function at node j

$$\phi_j^{(1)}(x_k) = \delta_{jk} \; ; \; (\phi_j^{(1)})'(x_k) = 0$$

$$\phi_j^{(2)}(x_k) = 0 \; ; \; (\phi_j^{(2)})'(x_k) = \delta_{jk}$$

$\phi_j^{(1)}$ switches on the deflection at node j
$\phi_j^{(2)}$ switches on the slope at node j

→ trial function

$$U(x_k) = \sum_{j \neq} \xi_j \cdot \phi_j^{(1)}(x_k) + \sum_j \eta_j \phi_j^{(2)}(x_k) = \xi_k$$

$$U'(x_k) = \sum_j \xi_j (\phi_j^{(1)})'(x_k) + \sum_j \eta_j (\phi_j^{(2)})'(x_k) = \eta_k$$

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The basis ϕ

~~H.W-3~~

→ Node j carries two dofs. So it needs two basis function

	x_{j-1}	x_j	x_{j+1}
deflects node j by 1 without rotating it ← $\phi_j^{(1)}$	0	1	0
$(\phi_j^{(1)})'$	0	0	0
rotate node j by 1 without moving it ← $Q_j^{(2)}$	0	0	0
$(Q_j^{(2)})'$	0	1	0

both are zero at every other node, so each lives on only two elements at node j

→ FEM solution

$$U = \sum \xi_j \phi_j^{(1)} + \xi \cdot \sum n_j \phi_j$$

(ξ: dof, ϕ: function)

→ $U(x_k) = \xi_k \cdot 3_k$ → deflection

$U'(x_k) = n_k$ → slope (θ)

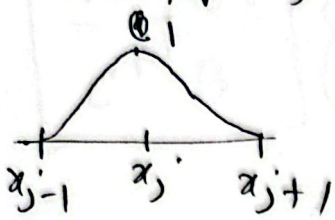
→ the explicit shape

$$\phi_j^{(1)} = \begin{cases} 3\left(\frac{x - x_{j-1}}{h}\right)^2 - 2\left(\frac{x - x_{j-1}}{h}\right)^3, & x_{j-1} \le x \le x_j \\ 3\left(\frac{x - x_j + 1}{+h}\right)^2 - 2\left(\frac{x - x_{j+1}}{-h}\right)^3, & x_j < x \le x_{j+1} \\ 0 & \text{otherwise} \end{cases}$$

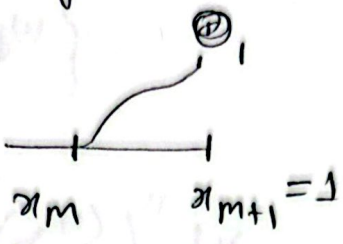
$$\phi_j^{(2)} = \begin{cases} -\frac{1}{h}(x - x_{j-1})^2 + \frac{1}{h^2}(x - x_{j-1})^3, & x_{j-1} \le x \le x_j \\ \frac{1}{h}(x - x_{j+1})^2 + \frac{1}{h^2}(x - x_{j+1})^3, & x_j \le x \le x_{j+1} \\ 0 & \text{otherwise} \end{cases}$$

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→ At the tip $j = M+1$ only the first branch exist.

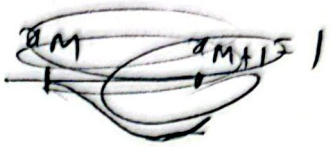


$\phi_j^{(1)}$ (interior)





→ half.



→ Trial function $U(x) = \sum_{j=1}^{M+1} \xi_j \phi_j^{(1)}(x) + \sum_{j=1}^{M+1} \eta_j \phi_j^{(2)}(x)$

$\xi_j := U(x_j)$

$\eta_j := U'(x_j)$

→ Test function $v = \phi_k^{(1)}, \phi_k^{(2)}$, when $k = 1, 2, \cdots M+1$

CG(3) FEM ~~Variational form~~

Find $U \in V_h$ such that $\int_0^1 U'' v'' dx = \int_0^1 f v \, dx$

for $\forall v \in V_h$

The linear system

substituting trial function and testing against each basis function gives $2(M+1)$ equation in the $2(M+1)$ unknows ξ, η. ~~Solve~~

$$\begin{bmatrix} A & C \\ C^T & B \end{bmatrix} \begin{Bmatrix} \alpha \\ \eta \end{Bmatrix} = \begin{Bmatrix} b^{(1)} \\ b^{(2)} \end{Bmatrix}$$

where, $A_{kj} = \int_0^1 (\phi_j^{(1)})'' (\phi_k^{(1)})'' \, dx$

$$B_{kj} = \int_0^1 (\phi_j^{(2)})'' (\phi_k^{(2)})'' \, dx$$

$$c_{kj} = \int_0^1 (\phi_j^{(2)})'' (\phi_k^{(1)})'' \, dx$$

→ on a uniform mesh this are ~~lecture 12 notes~~

$$A = \frac{12}{h^3} \begin{bmatrix} 2 & -1 & & & \\ 1 & 2 & \ddots & & \\ & \ddots & \ddots & \ddots & \\ & & \ddots & 2 & -1 \\ & & & -1 & \boxed{1} \end{bmatrix}; \quad B = \frac{2}{h} \begin{bmatrix} 4 & 1 & & & \\ 1 & 4 & \ddots & & \\ & \ddots & \ddots & \ddots & \\ & & \ddots & 4 & 1 \\ & & & 1 & \boxed{2} \end{bmatrix}$$

$\boxed{2}$ ← 4

$$c = \frac{6}{h^2} \begin{bmatrix} 0 & 1 & & & \\ -1 & 0 & \ddots & & \\ & \ddots & \ddots & \ddots & \\ & & \ddots & 0 & 1 \\ & & & -1 & \boxed{-1} \end{bmatrix}$$

$\boxed{-1}$ ← 0

∞ except · in the last row and column □

→ where node $n+1$ collects only one element instead of two.

load vector

$$b_k^{(1)} = \int_0^1 f \phi_k^{(1)} \, dx$$

$$b_k^{(2)} = \int_0^1 f \phi_k^{(2)} \, dx$$

$\Big\}$ $k = 1, 2 \ldots M+1$

~~Solution~~

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problem-9

prove a priori error estimate for this method in energy norm.

$$\|e\|_E = \left\{ \int_0^1 (e'')^2 \, dx \right\}^{1/2}$$

Solu

→ exact solution u satisfies for every $v \in V$

$$\int_0^1 u'' v'' \, dx = \int_0^1 f v \, dx \qquad \boxed{\forall v \in V}$$

→ FEM solution satisfies the same identity, but only against $v \in V_h$:

$$\int_0^1 U'' v'' \, dx = \int_0^1 f v \, dx \qquad \boxed{\forall v \in V_h}$$

→ $V_h \subset V$, so first line is also true for every $v \in V_h$. subtract for those $V-$ rhs are identical and cancel:

$$\int_0^1 e'' v'' \, dx = 0 \quad \text{for all } v \in V_h \quad \text{where } e = u - U$$

← Galerkin orthogonality

cauchy inequality

→ $\left| \int_0^1 a v' w' dx \right| \leq \|v\|_E \|w\|_E$

→ For this problem pairing is of second derivatives. with $a \equiv 1$, so the same inequality reads.

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$\left| \int_0^1 v'' w'' \, dx \right| \leq \|v\|_E \|w\|_E$ where

$$\|v\|_E = \left(\int_0^1 (v'')^2 dx \right)^{1/2}$$

$$\|w\|_E = \left(\int_0^1 (w'')^2 dx \right)^{1/2}$$

Step 3

– U is the best approximation in the energy norm

→ start from the definition, both factors e:

$$\|e\|_E^2 = \int_0^1 e'' e'' \, dx$$

→ pick any $v \in V_h$ and insert $-v+v$ in the second factor; $e = u - U = (u-v) + (v-U)$

$$\|e\|_E^2 = \int_0^1 e''(u-v)'' dx + \int_0^1 e''(v-U)'' dx$$

$\int_0^1 e''(v-U)'' dx \to 0$

→ so for any $v \in V_h$,

$$\|e\|_E^2 = \int_0^1 e''(u-v)'' dx$$

So, $\|e\|_E^2 \leq \|e\|_E \|u-v\|_E$

→ cancell one factor $\|e\|_E$ in the energy norm and (if it is zero, $U=u$ then is nothing to prove.

$$\|u-U\|_E \leq \|u-v\|_E \text{ for every } v \in V_h$$

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→ above holds for every $v \in V_h$

so, $\pi_h u = \sum_{j=1}^{M+1} u(x_j)\,\phi_j^{(1)} + \sum_{j=1}^{M+1} u'(x_j)\,\phi_j^{(2)}$

put $v = \pi_h u$

so, $\|u - U\|_E \le \|u - \pi_h u\|_E = \|(u - \pi_h u)''\|$

before 12 states the three hermite interpolation errors. the one that matches above. is the second derivative.

~~$\|u - U\|_E \le \|u - \pi_h u\|_E = (u - \pi_h u)''$~~

$$\|u - \pi_h u)''\|_{L^2(0,1)} \le c_3 \|h^2 u^{(4)}\|_{L^2(0,1)}$$

Now, $\|u - U\|_E \le c_3 \|h^2 u^{(4)}\|$

→ on a uniform mesh h is a constant. and comes from norm:

$$\|u - U\|_E \le c_3 h^2 \|u^{(4)}\| = O(h^2)$$

For CG(3) on cantilever, with $u^{(4)} = f$ · square integrable

$$\|u - U\|_E = \left\{ \int_0^1 \left((u - U)''\right)^2 dx \right\}^{1/2} \le c_3 h^2 \|u^{(4)}\|$$

→ energy norm converges at second order in h. halving the mesh devides error by 4

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problem-5

Statement

prove an a priori error estimate for this method in the L^2 norm.

$$\|e\|_{L^2} = \left\{ \int_0^1 (e)^2 \, dx \right\}^{1/2}$$

Solution

→ lecture 7 solves $-(a\phi')' = e$ for the second order problem. The beam operator is $\frac{d^4}{dx^4}$, so the same device ready →

$$\phi^{(iv)} = e \quad \text{on } (0,1)$$

$$\phi(0) = \phi'(0) = 0, \quad \phi''(1) = \phi'''(1) = 0 \quad \cdots (33)$$

→ same cantilever BCs as the original problem.

→ As in lecture (7) → this is legitimate bez problem is self adjoint. The dual operator is the same $(\cdot)^{(iv)}$, so we are not obliged to solve a different equation.

→ ϕ is never computed. It exists only to be paired against e,

→ Now, $\|e\|^2 = \int_0^1 ee \, dx = \int_0^1 e\,\phi^{(iv)} \, dx$.

$$= \underbrace{[e\phi''']_0^1}_{\to 0} - \underbrace{[e'\phi'']_0^1}_{\to 0} + \int_0^1 e''\phi'' dx$$

So, $\|e\|^2 = \int_0^1 e'' \phi'' \, dx$

Now, $\rightarrow \pi_n \phi \in V_n$ given, $\int_0^1 e''(\pi_n \phi)'' dx = 0$

Subtract. $\|e\|^2 = \int_0^1 e'' \phi'' dx - \underbrace{\int_0^1 e''(\pi_n \phi)'' dx}_{\rightsquigarrow 0}$

$$= \int_0^1 e''(\phi - \pi_n \phi)'' \, dx$$

$\rightarrow$ Now applying cauchy $\rightarrow$

$$\|e\|^2 \leq \|e''\| \, \|(\phi - \pi_n \phi)''\| = \|e\|_E \, \|(\phi - \pi_n \phi)''\|$$

$\rightarrow$ applying lecture 12 interpolation estimate,

$$\|(\phi - \pi_n \phi)''\| \leq c_3 h^2 \|\phi^{(4)}\|$$

Therefore, $\|e\|^2 \leq c_3 h^2 \|e\|_E \|e\|$

and cancelling one factor $\|e\|$

$$\|e\| \leq c_3 h^2 \|e\|_E$$

$\rightarrow$ From problem 4 $\Rightarrow$ $\|e\|_E \leq c_3 h^2 \|u^{(4)}\|$

So, $\|e\|_{L^2} \leq c_3^2 h^4 \|u^{(4)}\| = O(h^4)$

So, L^2 error converges two order faster than energy norm error

Problem 6

Numerically implement the CG(3) method on a uniform mesh with the following functions

$$\begin{aligned}
f(x) &= p_1(x)\cos(2\pi x) + p_2(x)\sin(2\pi x),\\
p_1(x) &= -64\pi(18 - 6\pi^4x^2 + 18x^3\pi^2 + 2\pi^4x - 36\pi^2x^2 - 27x + 12\pi^2x + 4\pi^4x^3 + 3\pi^2),\\
p_2(x) &= 216 - 816\pi^2 - 96\pi^4 + 32\pi^6x^2 - 384\pi^4x^3 - 64\pi^6x^3 - 288\pi^4x^2\\
&\quad + 3456\pi^2x + 576\pi^4x - 2592\pi^2x^2 + 144\pi^4x^4 + 32\pi^6x^4,\\
u(x) &= \big((18 + 2\pi^2)x^2 + (-24 - 4\pi^2)x^3 + (9 + 2\pi^2)x^4\big)\sin(2\pi x). \qquad (6.1)
\end{aligned}$$

Numerically determine the convergence rate in both the energy norm and the L^2-norm by creating an appropriate convergence table.

Step 1 — the data is consistent

- Checked symbolically before computing anything:

$$u^{(iv)} - f \equiv 0, \qquad u(0) = 0, \quad u'(0) = 0, \quad u''(1) = 0, \quad u'''(1) = 0.$$

- So (6.1) is a manufactured solution for exactly the cantilever of Problems 1–5, and u is available to measure the error against.

Step 2 — the implementation

- **Mesh.** N_e uniform elements, $h = 1/N_e$, nodes $x_i = ih$. In the lecture's labelling $M = N_e - 1$ interior nodes.
- **Unknowns.** Two per node. Node 0 is clamped, so both of its DOFs are deleted; the free tip keeps both. $\dim V_h = 2N_e$.
- **Stiffness.** Assembled element by element from the Hermite matrix with $EI = 1$ (Problem 3),

$$\boldsymbol{k}^e = \frac{1}{h^3}\begin{bmatrix} 12 & 6h & -12 & 6h \\ 6h & 4h^2 & -6h & 2h^2 \\ -12 & -6h & 12 & -6h \\ 6h & 2h^2 & -6h & 4h^2 \end{bmatrix},$$

  which reproduces the $\boldsymbol{A}, \boldsymbol{B}, \boldsymbol{C}$ blocks of Problem 3 exactly (checked against them directly).
- **Load.** $b_i = \int_0^1 f\phi_i \,\mathrm{d}x$ evaluated by 12-point Gauss–Legendre quadrature per element — *not* by interpolating f. f in (6.1) is a polynomial of degree ≤ 4 times $\sin$ / $\cos(2\pi x)$, so the rule is far more than enough; the results are identical to 9 digits with 8, 16 or 24 points.
- **Solve.** $\boldsymbol{S}\boldsymbol{d} = \boldsymbol{b}$ on the free DOFs.
- **Errors.** On each element, by 20-point Gauss,

$$\|e\|_E = \Big(\int_0^1 \big(u'' - U''\big)^2 \,\mathrm{d}x\Big)^{1/2}, \qquad \|e\|_{L^2} = \Big(\int_0^1 \big(u - U\big)^2 \,\mathrm{d}x\Big)^{1/2},$$

  with u'' differentiated in closed form (checked against a symbolic derivative to 5×10^{-13}).
- **Rate.** Successive meshes halve h, so rate $= \log_2\big(E(h)/E(h/2)\big)$.

Step 3 — convergence table

elements	M	h	$\|e\|_E$	rate	$\|e\|_{L^2}$	rate
4	3	0.25000	2.3314×10^{1}	—	6.4465×10^{-2}	—
8	7	0.12500	5.6259×10^{0}	2.05	3.9018×10^{-3}	4.05
16	15	0.06250	1.4162×10^{0}	1.99	2.4618×10^{-4}	3.99
32	31	0.03125	3.5490×10^{-1}	2.00	1.5434×10^{-5}	4.00
64	63	0.01562	8.8780×10^{-2}	2.00	9.6542×10^{-7}	4.00
128	127	0.00781	2.2199×10^{-2}	2.00	6.0369×10^{-8}	4.00

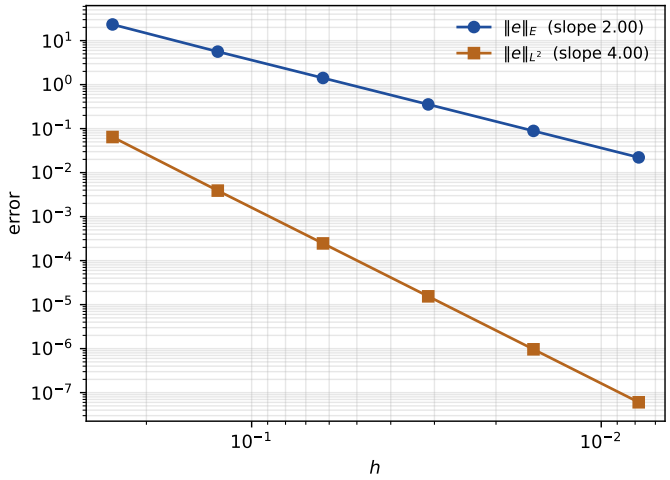


Figure 1: Convergence of the CG(3) cantilever solution under uniform refinement. Energy norm $O(h^2)$, L^2 norm $O(h^4)$.

Step 4 — the measured rates

- **Energy norm: rate $\to 2.00$.** This is the prediction of Problem 4, $\|e\|_E \le c_3 h^2 \|u^{(4)}\| = \mathcal{O}(h^2)$.
- **L^2 norm: rate $\to 4.00$.** This is the prediction of Problem 5, $\|e\|_{L^2} \le c_3^2 h^4 \|u^{(4)}\| = \mathcal{O}(h^4)$.
- The coarsest pair of meshes shows 2.05 and 4.05 — the asymptotic rate is not yet reached at $h = 1/4$, where the mesh barely resolves $\sin(2\pi x)$. From $h = 1/16$ down, both rates are clean to two decimals.
- Both estimates are therefore **sharp**: the observed rates are not merely $\ge$ the predicted ones, they equal them.

Conclusion. CG(3) on the cantilever converges at $\mathcal{O}(h^2)$ in the energy norm and $\mathcal{O}(h^4)$ in the L^2 norm, confirming Problems 4 and 5. The displacement converges two orders faster than the curvature, i.e. than the bending moment.